PassFrame tuition

Statistics & Averages

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 8

2. 17

3. 9

4. 11

5. 9

6. 15

7. 22

8. 35

9. A — 13

10. 29

11. 24 ((22+31+17+26)÷4 = 96÷4)

12. A, B and D (25 pupils were surveyed in total, The mode number of pets owned is 2, More pupils own 2 pets than own 3 pets)

13. A — 26

14. 9

15. 23

Standard

1. 10

2. 21

3. 44

4. A — 12

5. 37

6. 22 ((21+29+14+24)÷4 = 88÷4)

7. 7

8. 39 (45 pets originally, minus 6 = 39)

9. A and B (32 pupils were asked in total, Maths is the mode (most popular))

10. 15

11. A — 33

12. 9.5

13. 20

14. A — 24

15. 25

Standard / Hard

1. 26

2. 33

3. 20 (reading = 16, sport = 12, art = 48 − 16 − 12 = 20)

4. A — 9

5. 35

6. Mode = 19 (appears twice). Mean = (16+19+13+22+19+25)÷6 = 114÷6 = 19.

7. A, B, C and D (The mean daily sale is 54, The range of sales is 35, Thursday had the highest sales, Wednesday had the lowest sales)

8. 20

9. 72° (360 ÷ 5)

10. A — 34

11. 48

12. 16 (afternoon = 8, mean = (24+8)÷2 = 32÷2)

13. 45

14. 74 (ordered: 62, 66, 74, 78, 85)

15. A — 24

Homework — Answers

Easy / Standard

1. 10

2. 15

3. 11

4. 20

5. 10

6. 16

7. 14

8. 35

9. A — 14

10. 34

11. 28 ((26+35+21+30)÷4 = 112÷4)

12. A, B and D (28 pupils were surveyed in total, The mode number of pets owned is 2, More pupils own 2 pets than own 3 pets)

13. A — 28

14. 8

15. 23

Standard

1. 12

2. 23

3. 52

4. A — 14

5. 37

6. 24 ((23+31+16+26)÷4 = 96÷4)

7. 8

8. 60 (53 pets originally, plus 7 = 60)

9. A and B (38 pupils were asked in total, Maths is the mode (most popular))

10. 17

11. A — 33

12. 11.5

13. 22

14. A — 26

15. 25

Standard / Advanced

1. Total = mean × number of days = £65 × 5 = £325.

2. Largest = smallest + range = 12 + 7 = 19.

3. Total = 19 × 4 = 76. Sum of three known numbers = 56. Fourth number = 76 − 56 = 20.

4. Let yellow = x, green = 4x. x + 4x = 360 → 5x = 360 → x = 72°.

5. In an ordered list of 5 numbers, the median IS the 3rd (middle) value, so the missing number is 24.

6. Mean = (3+1+4+2+5+3)÷6 = 18÷6 = 3. Median: ordered 1,2,3,3,4,5 → (3+3)÷2 = 3. Mode = 3. All three measures equal 3.

7. Half = 30 walk. A fifth = 12 cycle. Rest = 60 − 30 − 12 = 18 travel by car.

8. A — 24 (total was 22×7=154, remove 10 to get 144, 144÷6=24)

9. 60 (Car + Bus = 180° out of 360° = half of 120 pupils)

10. A, B and C (The total of all 25 scores is 1,450, At least one score is below 58, The highest score minus the lowest score is 36)

11. Chocolate angle = 360 − 140 − 100 − 60 = 60°. Chocolate weight = 60/360 × 720g = 120g.

12. Total = 28 × 5 = 140. Sum of four known numbers = 112. Fifth number = 140 − 112 = 28.

13. A — The total of the 8 numbers is 96

14. Class A total = 20 × 1.5 = 30. Class B total = 20 × 0.9 = 18. Combined = 48 pets over 40 pupils = mean of 1.2.

15. Brown = 180/360 × 42 = 21. Blue = 120/360 × 42 = 14. Difference = 7.